In [ ]:

```python
import cv2           # Pragya Singhal , CU24220050
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Load a grayscale image (or convert a color image to grayscale)
# Note: Replace 'input_image.jpg' with the path to your image,
# or use a sample image built into OpenCV (e.g., cv2.samples.findFile('lena.jpg'
image_path = 'img.jpeg'

# Fallback: Create a synthetic test image with patterns and edges if file doesn'
try:
    original_image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
    if original_image is None:
        raise FileNotFoundError
except Exception:
    print("Loading default sample image or generating synthetic test pattern..."
    # Creating a synthetic image with shapes and noise for demonstration
    original_image = np.zeros((300, 300), dtype=np.uint8)
    cv2.rectangle(original_image, (50, 50), (150, 150), 200, -1)
    cv2.circle(original_image, (220, 220), 40, 150, -1)
    # Add some salt-and-pepper noise for testing median filter
    np.random.seed(42)
    noise = np.random.randint(0, 256, (300, 300), dtype=np.uint8)
    original_image = cv2.add(original_image, cv2.threshold(noise, 245, 255, cv2.

# Step 3: Low-Pass Filters (Smoothing & Noise Reduction)  # Pragya Singhal , CU2
# 3.1 Gaussian Blur (good for general noise reduction while preserving edges bet
gaussian_blur = cv2.GaussianBlur(original_image, (5, 5), 0)

# 3.2 Median Filter (excellent for impulse / salt-and-pepper noise)
median_filter = cv2.medianBlur(original_image, 5)

# 3.3 Average / Mean Filter (basic smoothing using a normalized box filter)    #
average_filter = cv2.blur(original_image, (5, 5))

# Step 4: High-Pass Filters (Edge Enhancement & Gradients)     # Pragya Singhal
# 4.1 Laplacian Filter (highlights regions of rapid intensity change)
laplacian = cv2.Laplacian(original_image, cv2.CV_64F)
laplacian_abs = cv2.convertScaleAbs(laplacian)

# 4.2 Sobel Filters (Horizontal and Vertical Gradients)
sobel_x = cv2.Sobel(original_image, cv2.CV_64F, 1, 0, ksize=3) # Horizontal edge
sobel_y = cv2.Sobel(original_image, cv2.CV_64F, 0, 1, ksize=3) # Vertical edges
sobel_x_abs = cv2.convertScaleAbs(sobel_x)
sobel_y_abs = cv2.convertScaleAbs(sobel_y)
sobel_combined = cv2.addWeighted(sobel_x_abs, 0.5, sobel_y_abs, 0.5, 0)

# Step 5: Displaying the Outputs using Matplotlib   # Pragya Singhal , CU2422005
plt.figure(figsize=(15, 10))

titles = [
    'Original Image', 'Gaussian Blur (LPF)', 'Median Filter (LPF)',
    'Average Filter (LPF)', 'Laplacian (HPF)', 'Sobel Combined (HPF)'
]
images = [
    original_image, gaussian_blur, median_filter,
    average_filter, laplacian_abs, sobel_combined
]
```

```
for i in range(6):
    plt.subplot(2, 3, i + 1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i], fontsize=12)
    plt.axis('off')

plt.tight_layout()
plt.show()
```

